DAY – 6 INTRODUCTION TO NETWORK DEVICES

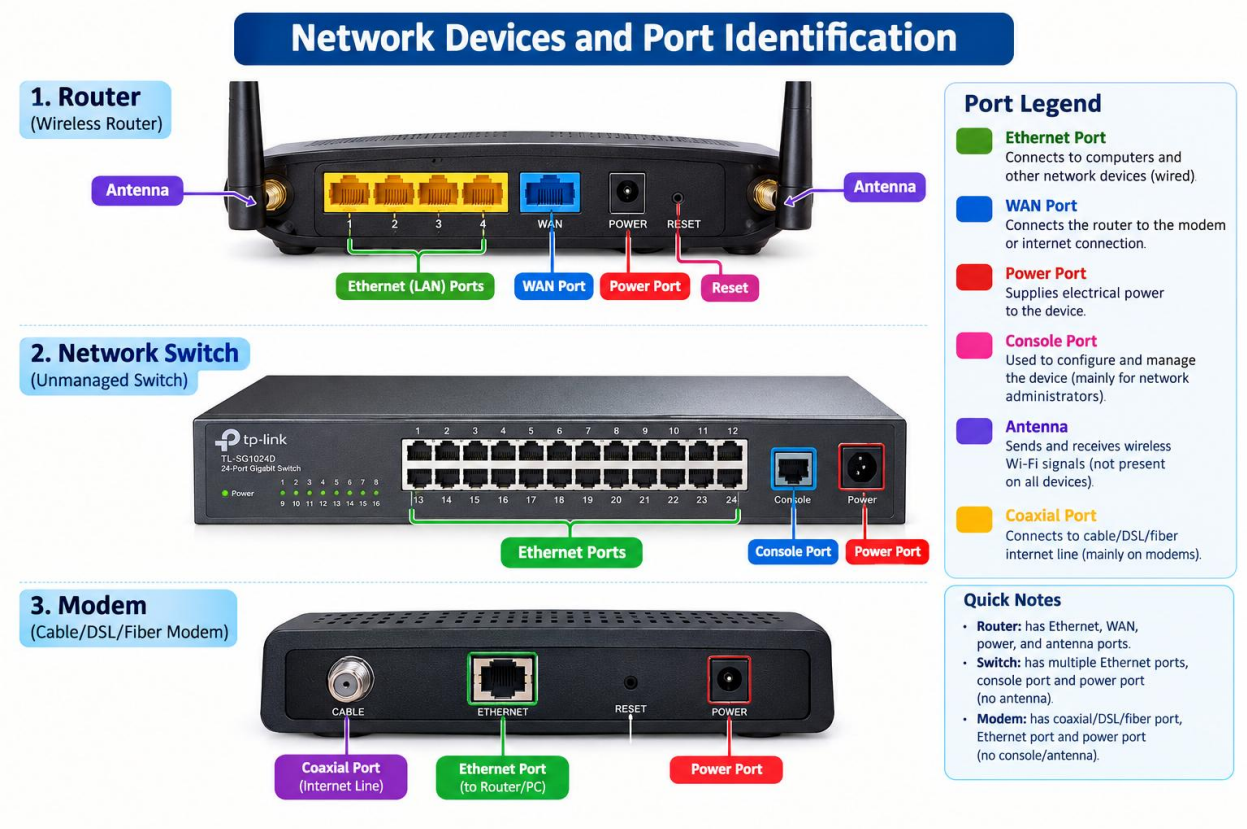
1. Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

Ans:-

Device	Main functions	Typical ports (home WiFi setup)
Switch	Connects wired devices (PC, printer, TV) in a LAN. It learns MAC addresses and sends data only to the correct port.	4 to 48 Ethernet ports, an uplink port to the router, power socket, and a console port on managed switches.
Router	Connects the home network to the Internet. Gives IP addresses (DHCP), uses NAT, chooses the path for data. Home routers usually include WiFi, a switch and a basic firewall.	1 WAN/Internet port, 4 LAN Ethernet ports, USB port, power socket, WiFi antennas, reset/WPS button.
Hub	Repeats the data it receives to all its ports. It is simple and slow, and is rarely used today.	4 to 8 Ethernet ports and a power socket.
Firewall	Checks network traffic and allows or blocks it using security rules, to protect the network from attacks.	WAN port, LAN port(s), DMZ port, console port, power socket. At home, it is usually built into the router.
Access Point	Creates the WiFi signal so wireless devices (Phones, laptops) can join the wired network.	1 Ethernet (LAN/PoE) port, power socket (DC), antennas, reset button.

2. Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, and antenna) for at least three devices.

Ans:-



* Router: Connects devices to the internet using Ethernet (LAN), WAN, power ports, and antennas.

* Network Switch: Connects multiple computers through Ethernet ports. Managed switches may also have a console port and power port.

* Modem: Connects to the internet service line through a coaxial, DSL, or fiber port and connects to a router through Ethernet.

3. Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

Ans:-

- App: Instagram

When I use Instagram, my phone connects to the internet through a Wi-Fi router or mobile network. The router forwards data between my local network and the internet, allowing me to open Instagram, watch reels, and upload photos. An access point provides a wireless connection between my phone and the local network using Wi-Fi. A firewall helps protect Instagram's servers by filtering network traffic and blocking unauthorized access. A network switch connects multiple servers and network devices within Instagram's data centers, helping them communicate and transfer data efficiently. All these network devices work together to provide a smooth, secure, and reliable Instagram experience.

4. Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Hint: Think about device capacity, coverage, and traffic handling.

Ans:-

- **Possible reason:** Too many students are using the same Wi-Fi network at the same time. The router may not be powerful enough to handle all connected devices, which can cause slow internet speeds and frequent disconnections. Weak Wi-Fi signals and interference from nearby networks can also affect connectivity.

- **Upgrade that can help:** Upgrade to a modern Wi-Fi 6 router or install additional access points to distribute the network load. This will improve network performance, provide better coverage, and allow more students to connect smoothly. The network switch should support Gigabit Ethernet to maintain high data-transfer speeds. The router should also have sufficient bandwidth to handle all users, and access points should be placed in suitable locations to reduce dead zones. Regular network monitoring can help identify and fix connectivity problems.